

Wetting boundary schemes in modified phase-field lattice Boltzmann method for binary fluids with large density ratios

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ABSTRACT

In this work, wetting boundary schemes in a modified phase-field lattice Boltzmann (LB) method are developed to simulate the contact-line motion and contact angle hysteresis of binary fluids with large density ratios. In the modified phase-field LB method, two multiple-relaxation-time (MRT) LB equations are utilized, one with a nondiagonal relaxation matrix used to solve the conserved Allen-Cahn (A-C) equation and the other with a modified discrete source term used to solve the incompressible Navier-Stokes (N-S) equations. Based on the idea of geometric formulation, two wetting boundary schemes corresponding to the modified phase-field LB model are proposed. In the first scheme, a layer of virtual nodes is utilized to realize the geometric formulation. The values of the order parameter at the virtual nodes are determined by the central difference method. In the second scheme, the geometric formulation is realized through a local scheme for the gradient terms of the order parameter where the nonequilibrium parts of the distribution function are utilized. Compared with the first wetting boundary scheme and some previous models, the second wetting boundary scheme does not require the virtual nodes and simplifies the numerical implementation. The developed two schemes are validated through a series of benchmark simulations, including droplet spreading on idea substrate, capillary intrusion and droplet shearing on nonideal substrate. It is found that satisfactory accuracy can be achieved by both two schemes, and the second scheme can obtain better stability while the first scheme shows better performance in simulating contact angle hysteresis.

1. Introduction

Wetting phenomena widespread in modern industrial applications, such as the spray coating process, the oil recovery process and droplet manipulation in microfluidic chips, etc., are receiving increasing attentions [1]. The contact-line dynamics, essentially as a complex multiphase flow, plays an important role in determination of the fluid behavior in wetting phenomena. In spite of the impressive progress for studying the contact-line dynamics with theoretical and experimental methods [2–4], numerical modeling is becoming an efficient and powerful tool to obtain better understanding of the wetting phenomena with the great advances in computer power [5–7]. In numerical methods, the wetting boundary scheme plays crucial role in determining the contact angle and accurately simulating the wetting phenomena. However, it is still a challenging problem to develop an accurate and efficient wetting boundary scheme with a powerful numerical model for multiphase flow especially with large density ratios. This is partially because of the complex interactions among different phases, wide range of length

scale from the macro size to the intermolecular distance and the large density change across the interface of liquid and gas phases. In this regard, the lattice Boltzmann (LB) method with the mesoscopic kinetic essences, easy boundary treatment and inherent parallelizable computational property has been considered as a powerful tool to simulate multiphase flow and wetting phenomena, during the past few decades [8–12]. Recently, the phase-field LB method, as one of the typical multiphase LB models, attracts many attention due to its simplicity and accuracy considered having great potential to simulate contact-line motion with large density ratios [13–16]. Therefore, developing effective wetting boundary schemes for the phase-field LB method is in need.

Generally, the wetting boundary schemes in the phase-field LB method fall into three categories: the fluid-solid interaction scheme, the surface-energy formulation scheme and the geometric formulation scheme. The fluid-solid interaction scheme is first proposed by Martys and Chen [17] and then further developed by Li et al. [18] in the pseudopotential LB model. In this model, the solid density is introduced to mimic the fluid-solid interaction and to control the contact angle. The

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major advantages of this method are that it can be easily extended to other models and is inherently suitable to simulate wetting phenomena in porous media [19]. Liu et al. [20] coupled the fluid-solid interaction scheme with a color-gradient LB model to simulate liquid CO₂ displacing water in a dual-permeability pore network. From a similar perspective, Huang et al. [21] developed an improved color-gradient LB method to investigate immiscible displacement in complex heterogeneous media for two-phase flows with kinematic viscosity contrast. Iwahara et al. [22] are one of the first to introduce the fluid-solid interaction scheme to the phase-field LB model. In their model, the solid density is determined by a so-called surface affinity which is related with the equilibrium contact angle. Davies et al. [23] replaced the solid density with the solid order parameter in the surface affinity to control the solid wettability. A similar contribution is also made by Yiotis et al. [24] base on the HCZ phase-field LB model [25]. In their work, a linear relationship is found between the equilibrium contact angle and the surface affinity. With the proposed model, Yiotis et al. [24] also studied the viscous coupling effects in immiscible two-phase flow in irregular pore networks. Although the fluid-solid interaction scheme is easy to implement, it also suffers from some problems, such as the extra requirement for prior simulations to determine the specific relationship between the solid density and the contact angle and the appearance of unphysical film caused by the nonzero gradient of density at the bulk phase [9,26,27]. The surface-energy formulation scheme is first proposed by Briant et al. [28,29] base on the free energy LB model. In their model, a linear wall free energy function is introduced to determine the wetting behavior of the fluid on a solid surface. Lee and Liu [30] is one of the first using the linear surface-energy formulation scheme into the phase-field LB model with a mixed difference scheme. Different from the fluid-solid interaction scheme, the contact angle can be prescribed in the surface-energy formulation scheme. However, adopting the linear surface-energy formulation will also result in the appearance of unphysical film [26,31,32]. To overcome it, several different forms of the surface-energy formulation are developed such as the sine form surface energy [33] and the cubic form surface energy [32,34,35]. Huang et al. [36] compared the performance of different surface-energy formulation in the phase-field LB model based on the Cahn-Hilliard (C-H) equation [37]. It is found in their work that the performances between the sine form surface energy and the cubic form surface energy are identical and both these two forms predict the wetting behavior more accurate than the linear form surface energy. Fakhari et al. [35] introduced the cubic form surface energy into the phase-field LB model based on the conserved Allen-Cahn (A-C) equation [38] to achieve the large density ratio and proposed an approximate method to apply the surface-energy formulation to curved boundaries. As for the third type of the wetting scheme, the geometric formulation scheme is proposed by Ding and Spelt [13]. They claimed that the geometric formulation and the cubic form surface-energy formulation are equivalent in computing the contact angle with a specific order parameter distribution while the geometric formulation is superior to the surface-energy scheme in correctly realizing a prescribed contact angle [13]. Liu et al. [39] introduce the geometric formulation into the color-gradient LB model. Then, according to the viewpoint in the geometric formulation, that is, the color gradient on the boundary is related to the geometry of the boundary, they extend this color-gradient model to simulate contact-line motion on curved boundaries [40,41]. Wang et al. [42] utilized the geometric formulation scheme into the phase-field LB model and extended it to simulate the contact angle hysteresis. Huang et al. [43] coupled the geometric formulation scheme with a hybrid phase-field LB model to study the motion of droplet on a surface with stepwise wettability gradient. Fakhri et al. [44] introduced the geometric formulation scheme into a phase-field LB model based on the conserved A-C equation to simplify the numerical implementation. Liang et al. [45] utilized an improved A-C based phase-field LB model with the geometric formulation wetting boundary scheme to simulate the contact-line motion of binary fluid, where extra temporal derivative is introduced to eliminate the er-

ror terms. With Liang et al.'s model [45], the wetting phenomena can be simulated under large density ratios. In Liang et al.'s work [45], the performance of the geometric formulation scheme is also compared with that of the cubic surface-energy formulation scheme. It is found that the geometric formulation scheme performs better than the surface-energy formulation scheme in simulating the contact angle hysteresis [45]. Recently, Begmohammadi et al. [46] claimed that in real applications, the extra temporal derivative utilized in Liang et al.'s model. [45] does not improve the accuracy obviously, but increases the difficulty in implementation. Note that, currently, a layer of virtual nodes is needed in all the aforementioned LB models to realize the geometric formulation [9,18,47].

In this work, wetting boundary schemes based on a modified phase-field LB model are developed to simulate the contact-line motion and the contact angle hysteresis of binary fluids with large density ratios on planar substrates. The behaviors of the binary fluid flows are described by the modified phase-field LB model, where two multiple-relaxation-time (MRT) LB equations are utilized. The first MRT LB equation is used to solve the conserved A-C equation with a nondiagonal relaxation matrix and corresponding equilibrium distribution function and discrete source term. The conserved A-C equation can be correctly recovered from the first MRT LB equation without the requirement for the extra temporal derivative. The second MRT LB equation is utilized to solve the Navier-Stokes (N-S) equations with a modified discrete force term. With this LB model, simulations for binary fluid flows with large density ratios can be achieved. Based on the idea of the geometric formulation, two corresponding wetting boundary schemes to the modified phase-field LB model are developed. The first one is similar with previous works, where the central difference method and a layer of virtual nodes are utilized to implement the geometric formulation. In the second one, using the nonequilibrium part of the distribution functions, a local scheme for the gradient terms of the order parameter is developed to realize the geometric formulation. Therefore, the virtual nodes are no longer needed in the second wetting boundary scheme. Both two wetting boundary schemes are validated and compared in the present work. The rest of the present paper is organized as follows. The modified phase-field LB model is introduced in Section 2. The wetting boundary schemes are developed in Section 3. The validation and comparisons of the two wetting boundary schemes are present in Section 4. Finally, Section 5 concludes the paper.

2. Modified phase-field LB model

In the modified phase-field LB model, two MRT LB equations with two distribution functions are utilized. The phase-field distribution function g is utilized to capture the phase interface and the pressure distribution function f is utilized to describe the velocity field. The two MRT LB equations are given by:

$$\begin{aligned} g(\mathbf{x} + \mathbf{e}\delta_t, t + \delta_t) - g(\mathbf{x}, t) \\ = \mathbf{M}^{-1} \left\{ -\Lambda_g \mathbf{M} [g(\mathbf{x}, t) - g^{\text{eq}}(\mathbf{x}, t)] + \left(\mathbf{I} - \frac{\Lambda_g}{2} \right) \delta_t \mathbf{R}_m \right\}, \end{aligned} \quad (1)$$

$$\begin{aligned} f(\mathbf{x} + \mathbf{e}\delta_t, t + \delta_t) - f(\mathbf{x}, t) \\ = \mathbf{M}^{-1} \left\{ -\Lambda_f \mathbf{M} [f(\mathbf{x}, t) - f^{\text{eq}}(\mathbf{x}, t)] + \left(\mathbf{I} - \frac{\Lambda_f}{2} \right) \delta_t \mathbf{F}_m \right\}, \end{aligned} \quad (2)$$

where $\mathbf{e}$ is the discrete velocity, δ_t is the time step, $\mathbf{M}$ is the transformation matrix, Λ_g is the relaxation matrix for the phase-field distribution function, Λ_f is the relaxation matrix for the pressure distribution function, g^{eq} and f^{eq} are the equilibrium phase-field distribution function and the equilibrium pressure distribution function, and $\mathbf{R}_m$ and $\mathbf{F}_m$ are the discrete source term and discrete force term.

For the sake of simplicity, the two-dimensional nine-velocity (D2Q9) lattice scheme [48] is utilized in the present work. Therefore, the discrete velocity in Eqs. (1) and (2) is given by

$$\mathbf{e} = \begin{cases} c(0,0)^T, & i=0, \\ c(\cos[(i-1)\pi/2], \sin[(i-1)\pi/2])^T, & i=1,2,3,4, \\ \sqrt{2}c(\cos[(2i-1)\pi/4], \sin[(2i-1)\pi/4])^T, & i=5,6,7,8, \end{cases} \quad (3)$$

where $c = \delta_x/\delta_t$ is the lattice speed, and δ_x is the lattice spacing. The transformation matrix is given by [49]

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -4 & -1 & -1 & -1 & -1 & 2 & 2 & 2 & 2 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{pmatrix}. \quad (4)$$

For the evolution equation of the phase-field distribution function described in Eq. (1), some modifications should be made to correctly recover the conserved A-C equation. First, inspired by Refs. [50–52] the relaxation matrix Λ_g is modified as a nondiagonal matrix described by

$$\Lambda_g = \begin{pmatrix} s_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & s_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & s_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & s_3 & s_4 \left(\frac{s_3}{2} - 1\right) & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & s_4 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & s_5 & s_6 \left(\frac{s_5}{2} - 1\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & s_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & s_7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & s_8 & 0 \end{pmatrix}, \quad (5)$$

where $s_0, s_1, \dots$, and s_8 are the relaxation factors. To be consistent with the nondiagonal relaxation matrix, the equilibrium distribution function is also modified as

$$\mathbf{n}^{\text{eq}} = \mathbf{M}\mathbf{g}^{\text{eq}} = \left(\phi, 2\phi, -2\phi, \frac{\phi u_x}{c}, -\frac{\phi u_x}{c}, \frac{\phi u_y}{c}, -\frac{\phi u_y}{c}, 0, 0 \right)^T, \quad (6)$$

where ϕ is the order parameter and $\mathbf{u} = (u_x, u_y)^T$ is the velocity calculated by Eq. (20a) below. The corresponding discrete source term $\mathbf{R}_m$ is given by

$$\mathbf{R}_m = \left(0, 0, 0, \frac{c[4\phi(1-\phi)]}{3W} \frac{\partial_x \phi}{|\nabla \phi|}, 0, \frac{c[4\phi(1-\phi)]}{3W} \frac{\partial_y \phi}{|\nabla \phi|}, 0, 0, 0 \right)^T, \quad (7)$$

where W is the phase interface thickness. The gradient terms of the order parameter $\nabla \phi$ involved in Eq. (7) is calculated through the isotropic central scheme, which is given by,

$$\nabla \phi = \frac{1}{c_s^2 \delta_t} \sum_i w_i \phi (\mathbf{x} + \mathbf{e}_i \delta_t) \mathbf{e}_i, \quad (8)$$

where $c_s = c/\sqrt{3}$ is the lattice sound speed and w_i is the weight coefficient related with the lattice scheme. In the D2Q9 lattice scheme, the weight coefficient is given by

$$w_i = \begin{cases} 4/9, & i=0, \\ 1/9, & i=1,2,3,4, \\ 1/36, & i=5,6,7,8. \end{cases} \quad (9)$$

Following the Chapman-Enskog analysis provided in Appendix A, the conserved A-C equation [38]

$$\partial_t \phi + \nabla \cdot (\phi \mathbf{u}) = \nabla \cdot \left[\mathcal{M} \left(\nabla \phi - \frac{4\phi(1-\phi)}{W} \frac{\nabla \phi}{|\nabla \phi|} \right) \right], \quad (10)$$

can be correctly recovered from the present LB model with no requirement for the temporal derivative utilized in the previous works

[14,45,53,54]. The mobility $\mathcal{M}$ and the order parameter ϕ are determined by

$$\phi = \sum_i g_i, \quad (11a)$$

$$\mathcal{M} = c_s^2 \delta_t \left(\frac{1}{s_{3,5}} - \frac{1}{2} \right). \quad (11b)$$

As for the evolution equation for the pressure distribution function $\mathbf{f}$ described in Eq. (2), a MRT version of the model proposed by Liang et al. [14] is developed and modified in the present work. The relaxation matrix Λ_f remains a diagonal matrix, which is given by:

$$\Lambda_f = \text{diag}(s_0, s_e, s_e, s_j, s_j, s_j, s_j, s_p, s_p). \quad (12)$$

The equilibrium pressure distribution function is given by

$$\mathbf{m}^{\text{eq}} = \mathbf{M}\mathbf{f}^{\text{eq}} = \begin{pmatrix} 0, \frac{3\rho|\mathbf{u}|^2 + 6p}{c^2}, -\frac{3\rho|\mathbf{u}|^2 + 9p}{c^2}, \frac{\rho u_x}{c}, -\frac{\rho u_x}{c}, \frac{\rho u_y}{c}, -\frac{\rho u_y}{c}, \frac{\rho(u_x^2 - u_y^2)}{c^2}, \frac{\rho u_x u_y}{c^2} \end{pmatrix}^T, \quad (13)$$

where p is the pressure and ρ is the density. In the phase-field model, the density is determined by

$$\rho = \phi(\rho_1 - \rho_2) + \rho_2, \quad (14)$$

where ρ_1 and ρ_2 are the density of the two fluids, respectively.

The discrete force term in Eq. (2) is modified in the present work as

$$\mathbf{F}_m = \left(\mathbf{u} \cdot \nabla \rho, \frac{6\mathbf{F} \cdot \mathbf{u}}{c^2}, -\mathbf{u} \cdot \nabla \rho, \frac{F_x}{c}, -\frac{F_x}{c}, \frac{F_y}{c}, -\frac{F_y}{c}, 2 \left[u_x \left(\frac{F_x}{c^2} + \frac{\partial_x \rho}{3} \right) - u_y \left(\frac{F_y}{c^2} + \frac{\partial_y \rho}{3} \right) \right], u_x \left(\frac{F_y}{c^2} + \frac{\partial_y \rho}{3} \right) + u_y \left(\frac{F_x}{c^2} + \frac{\partial_x \rho}{3} \right) \right)^T, \quad (15)$$

where $\mathbf{F} = \mathbf{F}_s + \mathbf{F}_b$ is the total force, $\mathbf{F}_s$ is the surface tension force, and $\mathbf{F}_b$ is the body force. Note that, compared with the discrete force term utilized in Liang et al.'s work [14], some extra terms in $\mathbf{u}\mathbf{F}$ are added in Eq. (15) to eliminate some error terms. Considering Eq. (14), the gradient term of the density in Eq. (15) can be determined by

$$\nabla \rho = (\rho_1 - \rho_2) \nabla \phi. \quad (16)$$

The surface tension force is given by [14,55]

$$\mathbf{F}_s = \mu_\phi \nabla \phi, \quad (17)$$

where μ_ϕ is the chemical potential as a function of the order parameter,

$$\mu_\phi = 4\beta\phi(\phi-1)(\phi-0.5) - \kappa\nabla^2\phi, \quad (18)$$

where β and κ are two parameters related with interface thickness W and the surface tension σ by $\beta = 12\sigma/W$ and $\kappa = 3\sigma W/2$.

As the Chapman-Enskog analysis made in Appendix B, the incompressible N-S equation

$$\nabla \cdot \mathbf{u} = 0, \quad (19a)$$

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot [\nu \rho (\nabla \mathbf{u} + \mathbf{u} \nabla)] + \mathbf{F}, \quad (19b)$$

can be recovered by the present LB model. The pressure p , the velocity $\mathbf{u}$, and the kinetic viscosity ν are determined by

$$\rho \mathbf{u} = \sum_i \mathbf{e}_i f_i + \frac{\delta_t}{2} \mathbf{F}, \quad (20a)$$

$$p = \frac{c_s^2}{1 - w_0} \left[\sum_{i \neq 0} f_i + \frac{\delta_t}{2} \mathbf{u} \cdot \nabla p - w_0 \frac{\rho \mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right], \quad (20b)$$

$$v = c_s^2 \delta_t \left(\frac{1}{s_p} - \frac{1}{2} \right). \quad (20c)$$

3. Wetting boundary scheme

When the two-phase fluids encounter with a solid substrate, the interface dynamics are affected by the contact-line motion determined by the wettability of the solid substrate. In numerical methods, the wettability of the solid substrate is reflected from the contact angle which is realized by wetting boundary schemes. From the geometric definition of the contact angle, Ding and Spelt [13] proposed a formulation which can be described by

$$\mathbf{n}_s \cdot \mathbf{n} = -\cos \theta, \quad (21)$$

where $\mathbf{n}_s$ is the normal vector to the interface, $\mathbf{n}$ is the normal vector to the solid substrate, and θ is the contact angle. In the diffuse-interface method, $\mathbf{n}_s$ can be determined by

$$\mathbf{n}_s = \frac{\nabla \phi}{|\nabla \phi|}. \quad (22)$$

Substituting Eq. (22) to Eq. (21), it can be obtained that

$$\mathbf{n} \cdot \nabla \phi = -|\nabla \phi| \cos \theta. \quad (23)$$

Through Eq. (23), the contact angle with a specific order parameter distribution can be computed. Ding and Spelt [13] also proved that in phase-field method Eq. (23) is equivalent to the cubic surface-energy formulation in determining the contact angle.

Note that in real applications, instead of computing the contact angle of a specific order parameter distribution, the main concerning is realizing the wettability of a solid substrate through a prescribed contact angle. In this regard, Ding and Spelt [13] supposed that the magnitude of the tangential component of $\nabla \phi$ (i.e., $\tau \cdot \nabla \phi$, where τ denotes the direction parallel to the solid substrate) should not be changed. Therefore, the prescribed contact angle should be realized by adjusting $\mathbf{n} \cdot \nabla \phi$, which is given by

$$\mathbf{n} \cdot \nabla \phi = -\tan \left(\frac{\pi}{2} - \theta \right) |\tau \cdot \nabla \phi|. \quad (24)$$

Eq. (24) is the so-called geometric formulation. Ding and Spelt [13] claimed that with Eq. (24) the prescribed contact angle can be achieved at each time step, while with the cubic surface-energy formulation, the realized contact angle will deviate from the prescribed contact angle.

In this section, with the idea of the geometric formulation, two wetting boundary schemes corresponding to the modified phase-field LB model introduced in Section 2 are proposed to simulate the contact-line motion of binary fluids with large density ratios on planar substrates. In the first scheme, the numerical implementation of the modified phase-field LB model is the same as discussed in Section 2. Considering the gradient term of the order parameter is calculated through the isotropic central scheme [i.e., Eq. (8)], to realize the geometric formulation [i.e., Eq. (24)] a layer of virtual nodes is required. As sketched in Fig. 1, a layer of virtual nodes is located at $y = 0$, the wetting solid substrate is located at $y = 1/2$, and $y = 1, 2$ represent the fluid layers. Note that, the solid substrate is not located on the grid nodes because the half-way bounce-back boundary condition widely used in LB models is utilized in the present work. Therefore, with the second-order central difference method, the discrete form of the geometric formulation Eq. (24) can be described by

$$\frac{\phi_{x,1} - \phi_{x,0}}{\delta_x} = -\tan \left(\frac{\pi}{2} - \theta \right) |\tau \cdot \nabla \phi|_{y=1/2}. \quad (25)$$

The tangential component of $\nabla \phi|_{y=1/2}$ on the solid substrate involved can be obtained through a second-order extrapolation scheme as

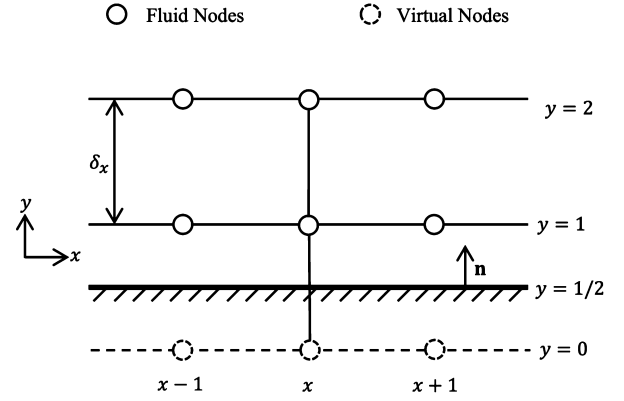


Fig. 1. Schematics of fluid nodes, solid substrate and the virtual nodes.

$$\tau \cdot \nabla \phi|_{y=1/2} = \frac{\partial \phi_{x,1/2}}{\partial x} = 1.5 \frac{\partial \phi_{x,1}}{\partial x} - 0.5 \frac{\partial \phi_{x,2}}{\partial x}. \quad (26)$$

The gradient term $\partial \phi / \partial x$ involved in Eq. (26) is also calculated through the second-order central difference method as

$$\frac{\partial \phi_{x,y}}{\partial x} = \frac{\phi_{x+1,y} - \phi_{x-1,y}}{2\delta_x}. \quad (27)$$

Note that the combination of Eqs. (25)–(27) is the first wetting boundary scheme in the present work to realize the geometric formulation. The key point of this method is determining the order parameter on the virtual nodes.

For the second wetting boundary scheme corresponding to the modified phase-field LB model, we attempt to avoid the requirement for the virtual nodes. It is clear that the geometric formulation [i.e., Eq. (24)] describes nothing but the relationship between the normal component and the tangential component of $\nabla \phi$ on the solid substrate. The tangential component of $\nabla \phi$ is maintained while the normal component of $\nabla \phi$ is forced to realize Eq. (24). Therefore, with a local scheme to calculate the gradient term $\nabla \phi$ the requirement for the virtual nodes can be avoided. In fact, local schemes for the gradient terms with distribution function are widely utilized in the LB models [11,44,54–57]. However, the local scheme in wetting boundary schemes hasn't been studied clearly. In this work, a local scheme to calculate $\nabla \phi$ with nonequilibrium parts of the phase-field distribution function is also proposed as

$$\nabla \phi = -\frac{\mathbf{A}}{|\mathbf{A}|} (B + |\mathbf{A}|), \quad (28)$$

where $\mathbf{A}$ and B are defined as

$$\mathbf{A} = \left(\frac{3s_3}{c\delta_t} \left(n_3^{(1)} + \frac{s_4}{2} n_4^{(1)} \right), \frac{3s_5}{c\delta_t} \left(n_5^{(1)} + \frac{s_6}{2} n_6^{(1)} \right) \right)^T, \quad (29a)$$

$$B = \left(1 - \frac{s_3}{2} \right) \frac{4\phi(1-\phi)}{W}, \quad (29b)$$

where $\mathbf{n}^{(1)} \approx \mathbf{Mg} - \mathbf{n}^{\text{eq}}$. With Eq. (28) the tangential component of $\nabla \phi$ on the solid substrate can be calculated locally. Then the normal component of $\nabla \phi$ can be determined by the geometric formulation Eq. (24). Note that, since the solid substrate is not located on the grid nodes, it is assumed in this method that the gradient term $\nabla \phi$ at the first layer of the fluid nodes (i.e., $y = 1$ in Fig. 1) also satisfies the geometric formulation Eq. (24). Besides, with the local scheme Eq. (28) the calculation of the velocity [i.e., Eq. (20a)] becomes implicit. Inspired by our previous work [11], the gradient term $\nabla \phi$ involved in Eq. (20a) is approximated by its value at the previous time step to avoid iteration. The applicability and accuracy of these two assumptions will be numerically validated in Section 4. Note that Eq. (28) is the key point of the second wetting boundary scheme in the present work to realize the geometric formulation. The main advantage of this scheme is that the efforts to determine the order parameter at the virtual nodes which could be complicated at some circumstance are no longer needed.

The two wetting boundary schemes developed above are utilized to simulate the wettability of a solid substrate by realizing a specific contact angle. However, in many real fluid-solid systems, the solid substrates are chemically inhomogeneous, which means the contact angle will vary over a range, i.e., the contact angle hysteresis window. The contact angle hysteresis also has a great effect on the contact-line motion. Therefore, simulating contact angle hysteresis is also a main purpose of wetting boundary schemes. In LB models, the contact angle hysteresis is usually determined through the prescription of a receding contact angle θ_R and an advancing contact angle θ_A [42,45,47]. Therefore, two methods to implement the contact angle hysteresis corresponding to the two wetting boundary schemes introduced above are also developed here. For the first wetting boundary scheme, the tangential component $\tau \cdot \nabla \phi$ on the solid substrate is first approximated through Eq. (26). Then, the contact angle θ is determined reversely through Eq. (25). If $\theta \leq \theta_R$, the contact angle should be set as $\theta = \theta_R$; if $\theta \geq \theta_A$ the contact angle should be set as $\theta = \theta_A$; otherwise, θ remains unchanged. With the determined contact angle θ , the order parameter at virtual nodes should be determined through the first wetting boundary scheme.

As for the second wetting boundary scheme, the gradient term $\nabla \phi$ at $y = 1$ is first calculated by Eq. (28) and the initial value of contact angle θ is approximated reversely through Eq. (24). Then with a similar comparison process between θ and θ_R and θ_A in the first method, the expected value of θ can be determined. Finally, the normal component of $\nabla \phi$ at $y = 1$ can be determined through the second wetting boundary scheme.

4. Numerical tests

4.1. Droplet spreading on an ideal substrate with prescribed contact angle

In this section, simulations for droplet spreading on an ideal substrate are employed to validate the capability of the present wetting boundary schemes in realizing prescribed contact angle, through investigating the equilibrium shapes of the droplet. The density ratio is set as $\rho^* = \rho_1/\rho_2 = 1000$ (where $\rho_1 = 1000$ and $\rho_2 = 1$, respectively) and the viscosity ratio is set as $\mu^* = \mu_1/\mu_2 = 100$ (where $\mu_1 = 10$ and $\mu_2 = 0.1$, respectively), which are similar with the properties of the realistic water-air system. Note that the capability of the present LB model to simulate contact-line motion with large density ratio can also be validated in these simulations. Other parameters are fixed as $\sigma = 0.2$, $\mathcal{M} = 0.1$ and $W = 5$. A semicircular droplet with radius $R = 128$ is first deposited on the bottom substrate of numerical domain with $N_x \times N_y = 1536 \times 512$. The periodic boundary condition is employed in the x direction, the no-slip boundary condition is utilized at the top and bottom substrates and the wetting boundary schemes developed in Section 3 are imposed at the bottom substrate.

With different prescribed contact angles indicating the wettability of the bottom substrate, different equilibrium shapes of droplet can be obtained by the present LB model and the two wetting boundary schemes, which are shown in Fig. 2. It can be found that both wetting boundary schemes can well simulate the droplet spreading on a solid substrate with a large range of contact angle from hydrophilic substrate ($\theta < 90^\circ$) to hydrophobic substrate ($\theta > 90^\circ$). To further validate and compare the two developed wetting boundary conditions, the contact angles obtained from the simulations should be compared with prescribed contact angles. From the equilibrium shapes of the droplet, the numerically predicted contact angle can be measured through

$$\theta = \frac{\pi}{2} + \arcsin \frac{y_c}{R_e}, \quad (30)$$

where y_c and R_e are the y coordinate of circle center and the radius of the equilibrium shape of the droplet, respectively, which can be obtained through the least square method. Since the phase-field LB model is a diffuse-interface model, the position of the interface is chosen at

$\rho_0 = 1/2(\rho_1 + \rho_2)$. Note that, in previous works [39,45] the numerically predicted contact angle is determined by the droplet spreading length L and its height H . However, as discussed in Section 3, since the half-way bounce back boundary condition is utilized in the present work, the substrate is not located on the grid nodes. Therefore, the spreading length L cannot be directly measured in this work. The numerically predicted contact angles obtained from both wetting boundary schemes and its absolute errors are shown in Table 1. It can be found that satisfactory accuracy can be achieved by both two wetting boundary conditions while with the second scheme the super hydrophilic substrate ($\theta < 20^\circ$) can be successfully described indicating better numerical stability of the second wetting boundary scheme. What's more, in the range of prescribed contact angle from 20° to 110° , the results obtained from the second wetting boundary scheme are slightly more accurate than the results obtained from the first wetting boundary scheme, which demonstrates the practicability of the assumption that the geometric formulation [i.e., Eq. (24)] is also satisfied at $y = 1$. However, it can also be observed that with super hydrophobic substrate ($\theta \geq 150^\circ$), the absolute errors by the second scheme are obviously larger than the absolute errors by the first scheme. Note that, with the super hydrophobic substrate, sharp increase of the absolute errors can be observed from both two schemes. In this case, the droplet has a significant tendency to leave the wall. Due the diffuse interface model adopted in the present work, higher grid resolution is needed for more accurate result. Since the geometric formulation is not realized on the wall in the second wetting boundary scheme, in this situation the deviations of the contact angle predicted by the second wetting boundary scheme from the prescribed contact angle are more obvious.

4.2. Capillary intrusion

In Section 4.1, two wetting boundary schemes in the present work are validated through a steady benchmark. Note that, in the second wetting boundary scheme the gradient term of the order parameter in Eq. (20a) is approximated by its value at the previous time step. Therefore, the second scheme should be tested through an unsteady benchmark. In this section, capillary instruction as a typical wetting phenomenon is simulated to further validate and compare the two wetting boundary schemes. As sketched in Fig. 3, the capillary pipe with size of $L \times H = 400 \times 20$ is initially filled with the non-wetting fluid and the outside wetting fluid would then intrude into the tube. The total numerical domain is set as $N_x \times N_y = 800 \times 36$, where the capillary pipe occupies the region $200 \leq x \leq 600$ and the non-wetting fluid is placed at $240 \leq x \leq 750$. The density ratio is set as $\rho^* = \rho_1/\rho_2 = 1000$ (where $\rho_1 = 1000$ and $\rho_2 = 1$, respectively) and the viscosity ratio is set as $\mu^* = \mu_1/\mu_2 = 100$ (where $\mu_1 = 10$ and $\mu_2 = 0.1$, respectively). Other parameters are chosen as $\sigma = 0.2$, $\mathcal{M} = 0.1$, and $W = 5$. The no-slip boundary condition is applied on all the pipe walls, the two wetting boundary schemes are employed to the upper and lower boundary of the pipe, and all the remaining boundaries are set as periodic boundaries. If ignoring the gravity and inertial effects, the evolution of the phase interface can be described as [45,58]

$$\sigma \cos \theta = \frac{6}{H} [\mu_2 (L - z) + \mu_1 z] \frac{dz}{dt}, \quad (31)$$

where $z(t)$ indicates the position of the interface and the initial position is set as $z(0) = 40$.

In our simulations, four different prescribed contact angles $\theta = 15^\circ, 30^\circ, 45^\circ, 60^\circ$ are considered. The evolution of the interface obtained by two wetting boundary schemes is shown in Fig. 4. Similar with the results in the last section, the first scheme encounters numerical instability at small prescribed contact angle $\theta = 15^\circ$ while the results obtained by the second scheme can still successfully simulate the capillary instruction in this circumstance. As for the rest three prescribed contact angles, the results obtained by both two wetting boundary schemes agree well with the analytical results obtained from Eq. (31). It

Table 1

Comparisons between the prescribed contact angle and the numerically predicted contact angle of two developed wetting boundary schemes.

Prescribed contact angles	Numerical contact angles		Absolute errors	
	First scheme	Second scheme	First scheme	Second scheme
12°	Divergence	12.0°	Divergence	0.0°
15°	Divergence	14.8°	Divergence	0.2°
20°	23.7°	22.3°	3.7°	2.3°
30°	29.8°	29.8°	0.2°	0.2°
40°	39.6°	39.7°	0.4°	0.3°
50°	50.4°	50.2°	0.4°	0.2°
60°	60.7°	60.3°	0.7°	0.3°
70°	70.5°	70.2°	0.5°	0.2°
80°	80.2°	80.0°	0.2°	0.0°
90°	89.8°	90.0°	0.2°	0.0°
100°	99.5°	100.2°	0.5°	0.2°
110°	109.4°	110.3°	0.6°	0.3°
120°	119.7°	120.8°	0.3°	0.8°
130°	130.2°	131.1°	0.2°	1.1°
140°	140.3°	140.8°	0.3°	0.8°
150°	151.2°	155.4°	1.2°	5.5°
160°	164.3°	167.9°	4.3°	7.9°

can also be found that identical results are obtained from the two wetting boundary schemes with $\theta = 30^\circ$ and 45° , while the results obtained by the second scheme are slightly closer to the analytical results than the results obtained by the first scheme with $\theta = 60^\circ$. Therefore, the assumption that the gradient term of the order parameter can be approximated by its value at the previous time step is practicable in real applications. Note that, with the increase of the contact angle, the capillary force determined by $\sigma \cos \theta$ decreases. Therefore, due to the factors not considered in Eq. (31) such as the inertial effect, initial-condition effect and inlet effect [58], the deviation of the numerical results obtained by the present model from the analytical results obtained by Eq. (31) can be observed under larger contact angle.

4.3. Droplet shearing on a nonideal substrate

In the last two benchmark simulations, the capability of the phase-field LB model with two developed wetting boundary schemes to simulate the contact-line motion of binary fluid with large density ratio on solid substrate with prescribed contact angles is well demonstrated. The practicability of the two assumptions for the second scheme introduced in Section 2 is also well proven. In this section, a droplet shearing on a nonideal substrate is investigated to test the ability of the wetting boundary schemes to simulate the contact angle hysteresis. Note that, in previous LB models [42,45,47] it is difficult to simulate the contact angle hysteresis under large density ratio. In the present work, the density ratio is set as $\rho^* = \rho_1/\rho_2 = 1000$ (where $\rho_1 = 1000$ and $\rho_2 = 1$, respectively) and the viscosity ratio is set as $\mu^* = \mu_1/\mu_2 = 100$ (where $\mu_1 = 10$ and $\mu_2 = 0.1$, respectively). The surface tensor is set as $\sigma = 5e^{-4}$. The numerical domain is set as $N_x \times N_y = 640 \times 128$, in which a semicircular droplet with radius $R = 64$ is initially placed on the bottom substrate centered at the node $(x_c, y_c) = (0.3N_x, 0)$. The periodic boundary condition is applied along the x direction, the no-slip boundary condition and the two wetting boundary schemes are employed on the bottom substrate, and the top substrate is set as a moving wall with a constant velocity $U_0 = 0.02$.

Typically, in this benchmark, four different motion models corresponding to four different hysteresis windows (θ_R, θ_A) are expected [42,45,47]. The first hysteresis window is $(\theta_R, \theta_A) = (0^\circ, 180^\circ)$. In this circumstance, both upstream and downstream dynamic contact angles cannot be beyond the hysteresis window. Therefore, the two contact points will be pinned on the surface. The second hysteresis window is $(\theta_R, \theta_A) = (0^\circ, 110^\circ)$. The downstream dynamic contact angle will be in excess of 110° with the shear force and the downstream contact point will slip on the substrate. Similarly, for the third hysteresis window $(\theta_R, \theta_A) = (70^\circ, 180^\circ)$, the upstream contact point will slip on the sub-

strate with the drive of the shear force. The last hysteresis window is $(\theta_R, \theta_A) = (70^\circ, 110^\circ)$. In this circumstance both the upstream and downstream contact points will slip along the substrate. In our simulation, all the four typical hysteresis windows are considered for both two wetting boundary schemes. The evolution of the interface predicted by the first wetting boundary scheme is shown in Fig. 5 and the evolution of the interface predicted by the second wetting boundary scheme is shown in Fig. 6. It can be found that all the four typical motion models with large density ratio can be correctly predicted by the first wetting boundary scheme. As for the second wetting boundary scheme, the third and fourth typical models [i.e., $(\theta_R, \theta_A) = (70^\circ, 180^\circ)$ and $(\theta_R, \theta_A) = (70^\circ, 110^\circ)$] can be successfully simulated. However, for the first and second typical models [i.e., $(\theta_R, \theta_A) = (0^\circ, 180^\circ)$ and $(\theta_R, \theta_A) = (0^\circ, 110^\circ)$] predicted by the second wetting scheme, slight sliding of the upstream contact point can be observed. To conclude, both two wetting boundary schemes can obtain satisfactory results for simulating the contact angle hysteresis with large density ratio, while the first wetting boundary scheme has better performance in this regard.

5. Conclusion

In this work, two wetting boundary schemes corresponding to a modified phase-field LB model are developed to simulate contact-line motion and contact angle hysteresis of binary fluids with large density ratios. In the modified phase-field LB model, two modified MRT LB equations are utilized, one for solving the conserved A-C equation and the other for solving the incompressible N-S equations. In the LB equation for the conserved A-C equation, a nondiagonal relaxation matrix is utilized to eliminate the error terms. In the LB equation for the N-S equations, a modified discrete force term is developed. Based on the idea of geometric formulation, two wetting boundary schemes are developed. In the first scheme, similar with the previous works, a lay of virtual nodes is used to realize the geometric formulation. The values of density in the virtual nodes are determined through the central difference method. In the second scheme, a local scheme to calculate the gradient term of the order parameter is developed to realize the geometric formulation. Two assumptions are also made for the second wetting scheme. The first assumption is that the geometric formulation can also be satisfied at the first fluid layer and the second assumption is that the gradient term involved in Eq. (20a) can be approximated by its value at previous time step. Both assumptions are validated through numerical simulations. With the second scheme, the requirement for the virtual nodes involved in the first wetting boundary scheme can be avoided. Numerical simulations indicate that satisfactory accuracy can

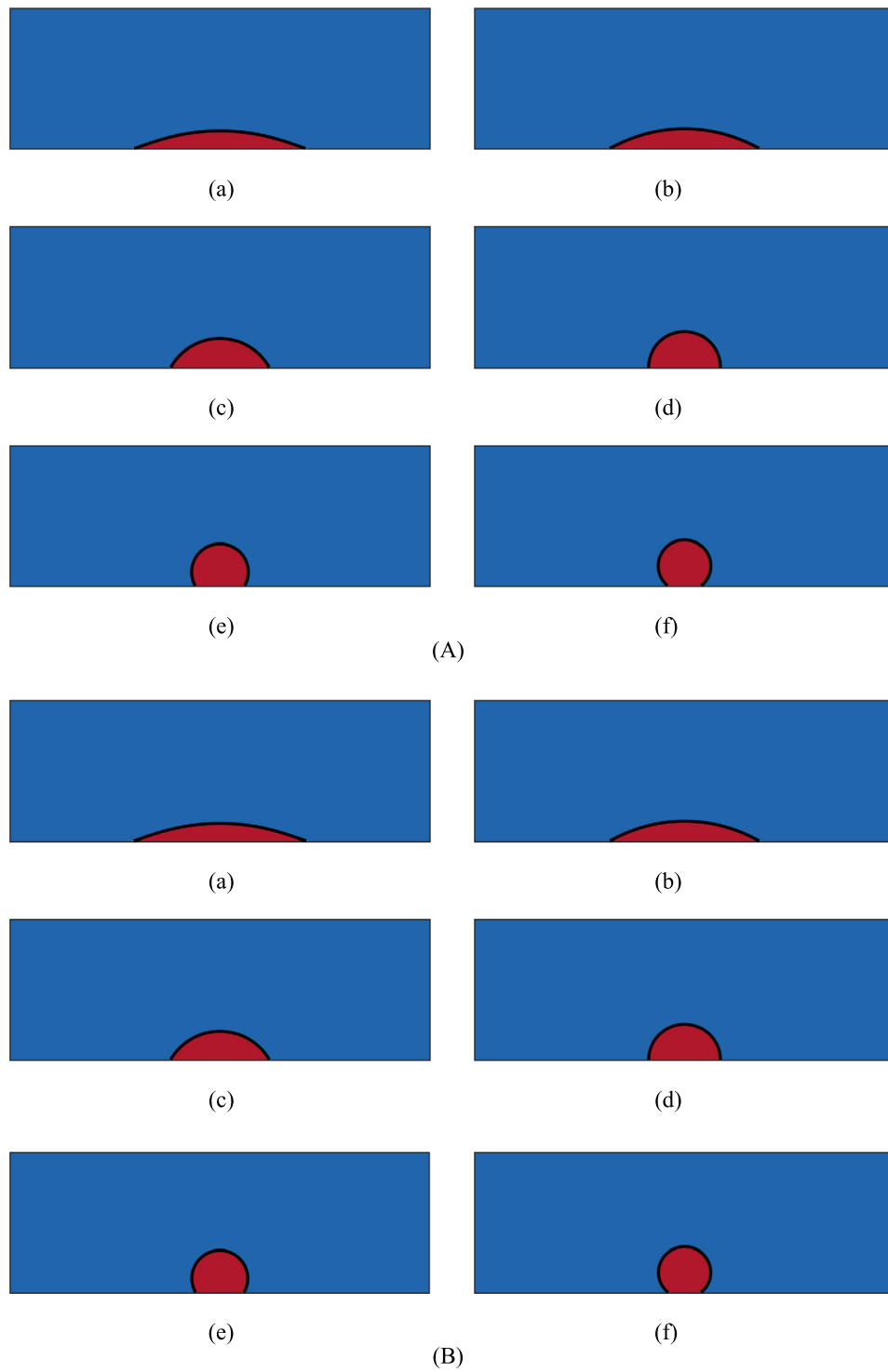


Fig. 2. The equilibrium shape of drop spreading on a substrate obtained through the first wetting boundary scheme (A) and the second wetting boundary scheme (B) with prescribed contact angles as 20° (a), 30° (b), 60° (c), 90° (d), 120° (e), and 140° (f).

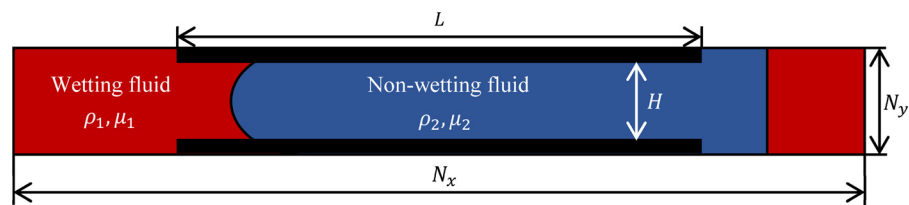


Fig. 3. Schematic of the capillary flow along a tube.

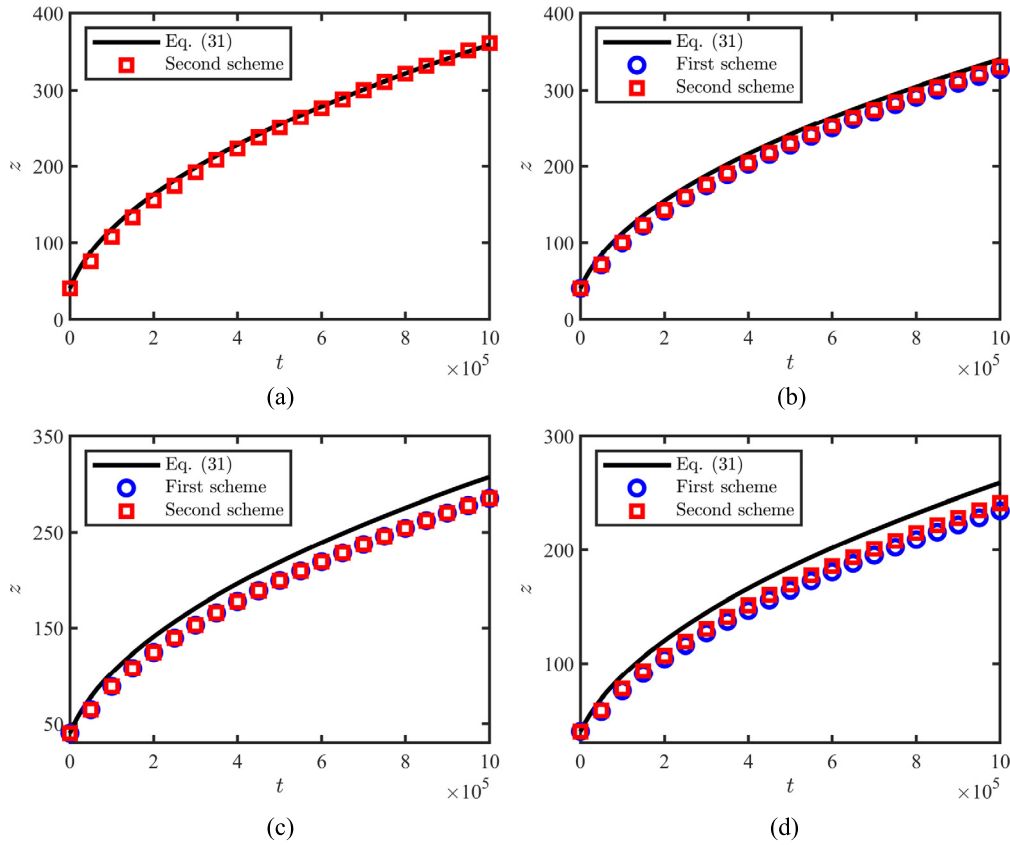


Fig. 4. Simulations of capillary instruction by the two wetting boundary schemes with the prescribed contact angles of $\theta = 15^\circ$ (a), $\theta = 30^\circ$ (b), $\theta = 45^\circ$ (c), and $\theta = 60^\circ$ (d).

be achieved by both two wetting boundary schemes to simulate contact-line motions on a substrate with prescribed contact angles. Larger range of contact angle and better stability can be obtained by the second wetting boundary scheme. The second scheme has better ability to simulate contact-line motion on a substrate with small contact angle ($\theta \leq 110^\circ$), while the first scheme is more suitable for super hydrophobic substrate ($\theta \geq 120^\circ$). The capability of the two developed wetting boundary schemes to simulate the contact angle hysteresis is validated through simulations for droplet shearing on a nonideal substrate. Four typical motion models can be basically predicted by two wetting boundary schemes while predictions by the first wetting boundary scheme are more precise.

Acknowledgements

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Appendix A. Chapman-Enskog analysis of Eq. (1)

Applying the Taylor series expansion to Eq. (1), we have

$$(\mathbf{I}\partial_t + \mathbf{D})\mathbf{n} + \frac{\delta_t}{2}(\mathbf{I}\partial_t + \mathbf{D})^2\mathbf{n} + O(\delta_t^2) = -\frac{\Lambda_g}{\delta_t}(\mathbf{n} - \mathbf{n}^{\text{eq}}) + \left(\mathbf{I} - \frac{\Lambda_g}{2}\right)\mathbf{R}_m, \quad (\text{A.1})$$

where $\mathbf{D} = \mathbf{M}[\text{diag}(\mathbf{e}_0 \cdot \nabla, \dots, \mathbf{e}_8 \cdot \nabla)]\mathbf{M}^{-1}$ and $\mathbf{n} = \mathbf{M}\mathbf{g}$. By introducing the following Chapman-Enskog expansions [54]

$$\begin{aligned} \partial_t &= \epsilon \partial_{t1} + \epsilon^2 \partial_{t2}, \quad \nabla = \epsilon \nabla_1, \\ \mathbf{n} &= \sum_{n=0}^{+\infty} \epsilon^n \mathbf{n}^{(n)}, \quad \mathbf{R}_m = \epsilon \mathbf{R}_m^{(1)}, \end{aligned} \quad (\text{A.2})$$

and substituting them into Eq. (A.1), Eq. (A.1) can be rewritten in the consecutive orders of the small expansion parameter ϵ :

$$\epsilon^0 : \mathbf{n}^{(0)} = \mathbf{n}^{\text{eq}}, \quad (\text{A.3a})$$

$$\epsilon^1 : (\mathbf{I}\partial_{t1} + \mathbf{D}_1)\mathbf{n}^{(0)} - \mathbf{R}_m = -\frac{\Lambda_g}{\delta_t} \left(\mathbf{n}^{(1)} + \frac{\delta_t}{2} \mathbf{R}_m \right), \quad (\text{A.3b})$$

$$\epsilon^3 : \partial_{t2}\mathbf{n}^{(0)} + (\mathbf{I}\partial_{t1} + \mathbf{D}_1) \left(\mathbf{I} - \frac{\Lambda_g}{2} \right) \left(\mathbf{n}^{(1)} + \frac{\delta_t}{2} \mathbf{R}_m^{(1)} \right) = -\frac{\Lambda_g}{\delta_t} \mathbf{n}^{(2)}. \quad (\text{A.3c})$$

Considering Eq. (11a) and Eq. (A.2), it can be obtained that

$$\mathbf{n}_0^{(n)} = 0 \quad (\forall n \geq 1). \quad (\text{A.4})$$

Then according to Eq. (A.3b), Eq. (6) and Eq. (7), we have

$$\partial_{t1}\phi + \nabla \cdot (\phi \mathbf{u}) = 0, \quad (\text{A.5a})$$

$$\begin{aligned} \partial_{t1} \left(\frac{\phi u_x}{c} \right) + \partial_{x1} \left(\frac{\phi c}{3} \right) \\ = -\frac{s_3}{\delta_t} n_3^{(1)} + \left(1 - \frac{s_3}{2} \right) \left[\frac{s_4}{\delta_t} n_4^{(1)} + \frac{c[4\phi(1-\phi)]}{3W} \frac{\partial_{x1}\phi}{|\nabla\phi|} \right], \end{aligned} \quad (\text{A.5b})$$

$$-\partial_{t1} \left(\frac{\phi u_x}{c} \right) = -\frac{s_4}{\delta_t} n_4^{(1)}, \quad (\text{A.5c})$$

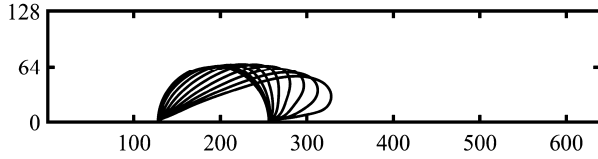
$$\begin{aligned} \partial_{t1} \left(\frac{\phi u_y}{c} \right) + \partial_{y1} \left(\frac{\phi c}{3} \right) = -\frac{s_5}{\delta_t} n_5^{(1)} + \left(1 - \frac{s_5}{2} \right) \left[\frac{s_6}{\delta_t} n_6^{(1)} + \frac{c[4\phi(1-\phi)]}{3W} \frac{\partial_{y1}\phi}{|\nabla\phi|} \right], \end{aligned} \quad (\text{A.5d})$$

$$-\partial_{t1} \left(\frac{\phi u_y}{c} \right) = -\frac{s_6}{\delta_t} n_6^{(1)}. \quad (\text{A.5e})$$

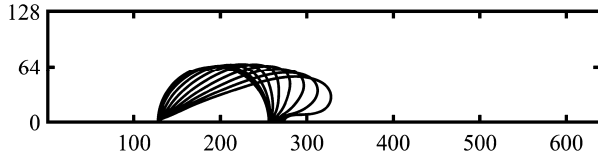
From Eq. (A.3c), it can be obtained that

$$\partial_{t2}\phi + c \nabla_1 \cdot \left(\left(1 - \frac{s_3}{2} \right) \left(n_3^{(1)} + \frac{s_4}{2} n_4^{(1)} + c \delta_t \frac{[4\phi(1-\phi)]}{6W} \frac{\partial_{x1}\phi}{|\nabla\phi|} \right) \right. \\ \left. \left(1 - \frac{s_5}{2} \right) \left(n_5^{(1)} + \frac{s_6}{2} n_6^{(1)} + c \delta_t \frac{[4\phi(1-\phi)]}{6W} \frac{\partial_{y1}\phi}{|\nabla\phi|} \right) \right) = 0. \quad (\text{A.6})$$

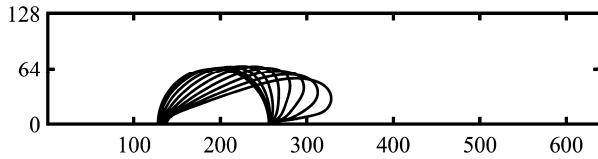
To simplify Eq. (A.6), the following relations are introduced by adding Eqs. (A.5b) and (A.5d) to Eqs. (A.5c) and (A.5e), respectively



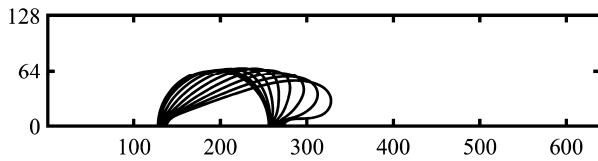
(a)



(b)



(c)



(d)

Fig. 5. The shearing of a droplet on a nonideal substrate predicted by the first wetting boundary scheme under different hysteresis windows as $(\theta_R, \theta_A) = (0^\circ, 180^\circ)$ (a), $(\theta_R, \theta_A) = (0^\circ, 110^\circ)$ (b), $(\theta_R, \theta_A) = (70^\circ, 180^\circ)$ (c), and $(\theta_R, \theta_A) = (70^\circ, 110^\circ)$ (d).

$$\partial_{x1} \left(\frac{\phi c}{3} \right) - \frac{c [4\phi(1-\phi)]}{3W} \frac{\partial_{x1}\phi}{|\nabla\phi|} = -\frac{s_3}{\delta_t} \left[n_3^{(1)} + \frac{s_4}{2} n_4^{(1)} + c\delta_t \frac{[4\phi(1-\phi)]}{6W} \frac{\partial_{x1}\phi}{|\nabla\phi|} \right], \quad (\text{A.7a})$$

$$\partial_{y1} \left(\frac{\phi c}{3} \right) - \frac{c [4\phi(1-\phi)]}{3W} \frac{\partial_{y1}\phi}{|\nabla\phi|} = -\frac{s_5}{\delta_t} \left[n_5^{(1)} + \frac{s_6}{2} n_6^{(1)} + c\delta_t \frac{[4\phi(1-\phi)]}{6W} \frac{\partial_{y1}\phi}{|\nabla\phi|} \right]. \quad (\text{A.7b})$$

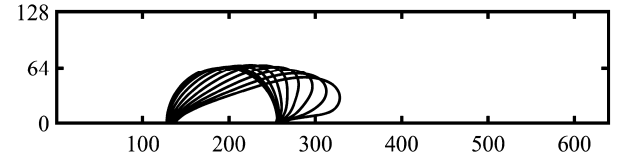
Substituting Eq. (A.7) into Eq. (A.6), it can be obtained that

$$\partial_{t2}\phi = \frac{c^2\delta_t}{3} \nabla_1 \cdot \left(\left(\frac{1}{s_3} - \frac{1}{2} \right) \left(\partial_{x1}\phi - \frac{[4\phi(1-\phi)]}{W} \frac{\partial_{x1}\phi}{|\nabla\phi|} \right) \left(\frac{1}{s_5} - \frac{1}{2} \right) \left(\partial_{y1}\phi - \frac{[4\phi(1-\phi)]}{W} \frac{\partial_{y1}\phi}{|\nabla\phi|} \right) \right). \quad (\text{A.8})$$

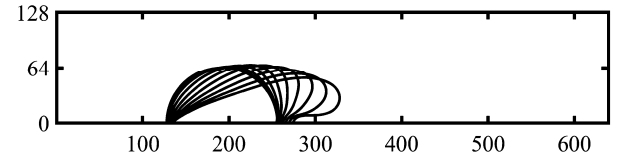
Adding Eq. (A.8) to Eq. (A.5a) and considering Eq. (11b) it can be obtained that

$$\partial_t\phi + \nabla \cdot (\phi \mathbf{u}) = \nabla \cdot \left[\mathcal{M} \left(\nabla\phi - \frac{4\phi(1-\phi)}{W} \frac{\nabla\phi}{|\nabla\phi|} \right) \right]. \quad (\text{A.9})$$

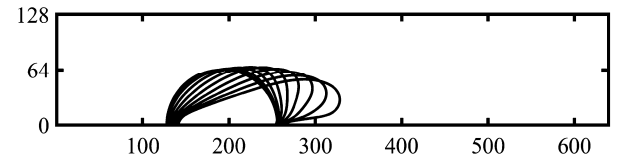
It is obvious that Eq. (A.9) is the same as Eq. (10) i.e., the conserved A-C equation. Therefore, the conserved A-C equation can be correctly recovered from Eq. (1) with no deviation term.



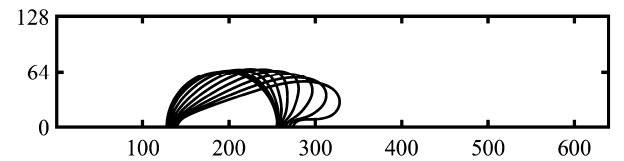
(a)



(b)



(c)



(d)

Fig. 6. The shearing of a droplet on a nonideal substrate predicted by the second wetting boundary scheme under different hysteresis windows as $(\theta_R, \theta_A) = (0^\circ, 180^\circ)$ (a), $(\theta_R, \theta_A) = (0^\circ, 110^\circ)$ (b), $(\theta_R, \theta_A) = (70^\circ, 180^\circ)$ (c), and $(\theta_R, \theta_A) = (70^\circ, 110^\circ)$ (d).

Then the local scheme developed in the second wetting boundary scheme [i.e. Eq. (28)] is derived. From Eq. (A.7), the symbol $\mathbf{A}$ can be written as

$$\mathbf{A} = \begin{pmatrix} \left[\left(1 - \frac{s_3}{2} \right) \frac{[4\phi(1-\phi)]}{W|\nabla\phi|} - 1 \right] \partial_x\phi \\ \left[\left(1 - \frac{s_5}{2} \right) \frac{[4\phi(1-\phi)]}{W|\nabla\phi|} - 1 \right] \partial_y\phi \end{pmatrix}. \quad (\text{A.10})$$

It is easy to derive from Eq. (A.10) that

$$\mathbf{A} = \left(\frac{B}{|\nabla\phi|} - 1 \right) \nabla\phi. \quad (\text{A.11})$$

Therefore, it can be obtained that

$$|\nabla\phi| = B \pm |\mathbf{A}|. \quad (\text{A.12})$$

To guarantee that $|\nabla\phi|$ is positive, it can be expected that

$$|\nabla\phi| = B + |\mathbf{A}|. \quad (\text{A.13})$$

Substituting Eq. (A.13) into Eq. (A.11), it can be obtained that

$$\nabla\phi = -\frac{\mathbf{A}}{|\mathbf{A}|} (B + |\mathbf{A}|), \quad (\text{A.14})$$

which is the same as Eq. (28), i.e., the local scheme developed in the second wetting boundary scheme.

Appendix B. Chapman-Enskog analysis of Eq. (2)

Applying the Taylor series expansion to Eq. (2), we have

$$(\mathbf{I}\partial_t + \mathbf{D})\mathbf{m} + \frac{\delta_t}{2}(\mathbf{I}\partial_t + \mathbf{D})^2\mathbf{m} + O(\delta_t^2) = -\frac{\Lambda_f}{\delta_t}(\mathbf{m} - \mathbf{m}^{\text{eq}}) + \left(\mathbf{I} - \frac{\Lambda_f}{2}\right)\mathbf{F}_m, \quad (\text{B.1})$$

where $\mathbf{m} = \mathbf{M}\mathbf{f}$. Similarly, the Chapman-Enskog expansion is introduced as

$$\begin{aligned} \partial_t &= \varepsilon \partial_{t1} + \varepsilon^2 \partial_{t2}, \nabla = \varepsilon \nabla_1, \\ \mathbf{m} &= \sum_{n=0}^{+\infty} \varepsilon^n \mathbf{m}^{(n)}, \mathbf{F}_m = \varepsilon \mathbf{F}_m^{(1)}, \end{aligned} \quad (\text{B.2})$$

and substituting them into Eq. (A.2), Eq. (A.2) can be rewritten in the consecutive orders of the small expansion parameter ε :

$$\varepsilon^0 : \mathbf{m}^{(0)} = \mathbf{m}^{\text{eq}}, \quad (\text{B.3a})$$

$$\varepsilon^1 : (\mathbf{I}\partial_{t1} + \mathbf{D}_1)\mathbf{m}^{(0)} - \mathbf{F}_m = -\frac{\Lambda_f}{\delta_t} \left(\mathbf{m}^{(1)} + \frac{\delta_t}{2} \mathbf{F}_m^{(1)} \right), \quad (\text{B.3b})$$

$$\varepsilon^2 : \partial_{t2}\mathbf{m}^{(0)} + (\mathbf{I}\partial_{t1} + \mathbf{D}_1) \left(\mathbf{I} - \frac{\Lambda_f}{2} \right) \left(\mathbf{m}^{(1)} + \frac{\delta_t}{2} \mathbf{F}_m^{(1)} \right) = -\frac{\Lambda_g}{\delta_t} \mathbf{m}^{(2)}. \quad (\text{B.3c})$$

As developed in Liang et al.'s work [14], it can be obtained that

$$m_0^{(1)} + \frac{\delta_t}{2} F_{m0}^{(1)} = 0, \quad (\text{B.4a})$$

$$m_3^{(1)} + \frac{\delta_t}{2} F_{m3}^{(1)} = 0, \quad (\text{B.4b})$$

$$m_5^{(1)} + \frac{\delta_t}{2} F_{m5}^{(1)} = 0, \quad (\text{B.4c})$$

$$m_{0,3,5}^{(n)} = 0 \quad (\forall n \geq 2). \quad (\text{B.4d})$$

According to Eq. (B.3b), it can be obtained that

$$\nabla_1 \cdot \mathbf{u} = 0, \quad (\text{B.5a})$$

$$\partial_{t1}(\rho u_x) + \partial_{x1}(\rho u_x^2) + \partial_{y1}(\rho u_x u_y) = -\partial_{x1}p + F_x, \quad (\text{B.5b})$$

$$\partial_{t1}(\rho u_y) + \partial_{x1}(\rho u_x u_y) + \partial_{y1}(\rho u_y^2) = -\partial_{y1}p + F_y. \quad (\text{B.5c})$$

Then it can be obtained from Eq. (B.3c) that

$$\begin{aligned} \partial_{t2}(\rho u_x) + \partial_{x1} \left[\frac{c^2}{6} \left(1 - \frac{s_e}{2} \right) \left(m_1^{(1)} + \frac{\delta_t}{2} F_{m1}^{(1)} \right) \right. \\ \left. + \frac{c^2}{2} \left(1 - \frac{s_p}{2} \right) \left(m_7^{(1)} + \frac{\delta_t}{2} F_{m7}^{(1)} \right) \right] \\ + \partial_{y1} \left[c^2 \left(1 - \frac{s_p}{2} \right) \left(m_8^{(1)} + \frac{\delta_t}{2} F_{m8}^{(1)} \right) \right] = 0, \end{aligned} \quad (\text{B.6a})$$

$$\begin{aligned} \partial_{t2}(\rho u_y) + \partial_{x1} \left[c^2 \left(1 - \frac{s_p}{2} \right) \left(m_8^{(1)} + \frac{\delta_t}{2} F_{m8}^{(1)} \right) \right] \\ + \partial_{y1} \left[\frac{c^2}{6} \left(1 - \frac{s_e}{2} \right) \left(m_1^{(1)} + \frac{\delta_t}{2} F_{m1}^{(1)} \right) \right. \\ \left. + \frac{c^2}{2} \left(1 - \frac{s_p}{2} \right) \left(m_7^{(1)} + \frac{\delta_t}{2} F_{m7}^{(1)} \right) \right] = 0. \end{aligned} \quad (\text{B.6b})$$

To simplify Eq. (B.6), some formulations still need to be derived from Eq. (B.3b) that

$$\frac{3\partial_{t1}(\rho u_x^2 + \rho u_y^2 + 2p)}{c^2} - \frac{6(F_x u_x + F_y u_y)}{c^2} = -\frac{s_e}{\delta_t} \left(m_1^{(1)} + \frac{\delta_t}{2} F_{m1}^{(1)} \right), \quad (\text{B.7a})$$

$$\begin{aligned} \frac{\partial_{t1}(\rho u_x^2 - \rho u_y^2)}{c^2} - \frac{2(F_x u_x - F_y u_y)}{c^2} + \frac{2}{3}\rho(\partial_{x1}u_x - \partial_{y1}u_y) \\ = -\frac{s_p}{\delta_t} \left(m_7^{(1)} + \frac{\delta_t}{2} F_{m7}^{(1)} \right), \end{aligned} \quad (\text{B.7b})$$

$$\frac{\partial_{t1}(\rho u_x u_y)}{c^2} - \frac{F_x u_y + F_y u_x}{c^2} + \frac{1}{3}\rho(\partial_{x1}u_y + \partial_{y1}u_x) = -\frac{s_p}{\delta_t} \left(m_8^{(1)} + \frac{\delta_t}{2} F_{m8}^{(1)} \right). \quad (\text{B.7c})$$

Before proceeding further, some statement should be made. As claimed in Ref. [59], the term $\partial_t p$ can be eliminated in low frequency limit and $\nabla p \sim O(\text{Ma}^3)$. Therefore, combining Eq. (B.7) and Eq. (B.5), Eq. (B.6) can be simplified as

$$\partial_{t2}(\rho u_x) = \partial_{x1}[\rho\nu(\partial_{x1}u_x - \partial_{y1}u_y)] + \partial_{y1}[\rho\nu(\partial_{x1}u_y + \partial_{y1}u_x)], \quad (\text{B.8a})$$

$$\partial_{t2}(\rho u_y) = \partial_{x1}[\rho\nu(\partial_{x1}u_y + \partial_{y1}u_x)] + \partial_{y1}[\rho\nu(\partial_{x1}u_x - \partial_{y1}u_y)]. \quad (\text{B.8b})$$

The $O(\text{Ma}^3)$ terms are eliminated in Eq. (B.8). Note that, in Liang et al.'s work [14], the terms in $\mathbf{F}_u$ are not included in the discrete force term. Therefore, these terms in Eq. (B.7) cannot be eliminated. Adding Eq. (B.8) into Eq. (B.5), it can be obtained that

$$\nabla \cdot \mathbf{u} = 0, \quad (\text{B.9a})$$

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \mathbf{F} + \nabla \cdot [\rho\nu(\nabla \mathbf{u} + \mathbf{u} \nabla)], \quad (\text{B.9b})$$

which are the same as the incompressible N-S equations described in Eq. (19).

References

- [1] J.H. Snoeijer, B. Andreotti, Moving contact lines: scales, regimes, and dynamical transitions, *Annu. Rev. Fluid Mech.* 45 (2013) 269–292, <https://doi.org/10.1146/annurev-fluid-011212-140734>.
- [2] D. Bonn, J. Eggers, J. Indekeu, J. Meunier, E. Rolley, Wetting and spreading, *Rev. Mod. Phys.* 81 (2009) 739–805, <https://doi.org/10.1103/RevModPhys.81.739>.
- [3] R.G. Cox, The dynamics of the spreading of liquids on a solid surface. Part 2. Surfactants, *J. Fluid Mech.* 168 (1986) 195, <https://doi.org/10.1017/S0022112086000344>.
- [4] R. Seemann, M. Brinkmann, T. Pfohl, S. Herminghaus, Droplet based microfluidics, *Rep. Prog. Phys.* 75 (2012) 016601, <https://doi.org/10.1088/0034-4885/75/1/016601>.
- [5] P.D.M. Spelt, A level-set approach for simulations of flows with multiple moving contact lines with hysteresis, *J. Comput. Phys.* 207 (2005) 389–404, <https://doi.org/10.1016/j.jcp.2005.01.016>.
- [6] B. Shi, V.K. Dhir, Molecular dynamics simulation of the contact angle of liquids on solid surfaces, *J. Chem. Phys.* 130 (2009) 034705, <https://doi.org/10.1063/1.3055600>.
- [7] D. Gueyffier, J. Li, A. Nadim, R. Scardovelli, S. Zaleski, Volume-of-fluid interface tracking with smoothed surface stress methods for three-dimensional flows, *J. Comput. Phys.* 152 (1999) 423–456, <https://doi.org/10.1006/jcph.1998.6168>.
- [8] Q. Li, K.H. Luo, Q.J. Kang, Y.L. He, Q. Chen, Q. Liu, Lattice Boltzmann methods for multiphase flow and phase-change heat transfer, *Prog. Energy Combust. Sci.* 52 (2016) 62–105, <https://doi.org/10.1016/j.peccs.2015.10.001>.
- [9] S. Wu, Y. Chen, L.-Q. Chen, Three-dimensional pseudopotential lattice Boltzmann model for multiphase flows at high density ratio, *Phys. Rev. E* 102 (2020) 053308, <https://doi.org/10.1103/PhysRevE.102.053308>.
- [10] Q. Li, Y. Yu, K.H. Luo, Implementation of contact angles in pseudopotential lattice Boltzmann simulations with curved boundaries, *Phys. Rev. E* 100 (2019) 053313, <https://doi.org/10.1103/PhysRevE.100.053313>.
- [11] S. Zhang, J. Tang, H. Wu, R. Huang, Improved thermal multiple-relaxation-time lattice Boltzmann model for liquid-vapor phase change, *Phys. Rev. E* 103 (2021) 043308, <https://doi.org/10.1103/PhysRevE.103.043308>.
- [12] F. Qin, L. Del Carro, A. Mazloomi Moqaddam, Q. Kang, T. Brunschweiler, D. Derome, J. Carmeliet, Study of non-isothermal liquid evaporation in synthetic micro-pore structures with hybrid lattice Boltzmann model, *J. Fluid Mech.* 866 (2019) 33–60, <https://doi.org/10.1017/jfm.2019.69>.
- [13] H. Ding, P.D.M. Spelt, Wetting condition in diffuse interface simulations of contact line motion, *Phys. Rev. E* 75 (2007) 046708, <https://doi.org/10.1103/PhysRevE.75.046708>.
- [14] H. Liang, J. Xu, J. Chen, H. Wang, Z. Chai, B. Shi, Phase-field-based lattice Boltzmann modeling of large-density-ratio two-phase flows, *Phys. Rev. E* 97 (2018) 033309, <https://doi.org/10.1103/PhysRevE.97.033309>.
- [15] Y.Q. Zu, A.D. Li, H. Wei, Phase-field lattice Boltzmann model for interface tracking of a binary fluid system based on the Allen-Cahn equation, *Phys. Rev. E* 102 (2020) 053307, <https://doi.org/10.1103/PhysRevE.102.053307>.
- [16] H.W. Zheng, C. Shu, Y.T. Chew, A lattice Boltzmann model for multiphase flows with large density ratio, *J. Comput. Phys.* 218 (2006) 353–371, <https://doi.org/10.1016/j.jcp.2006.02.015>.
- [17] N.S. Martys, H. Chen, Simulation of multicomponent fluids in complex three-dimensional geometries by the lattice Boltzmann method, *Phys. Rev. E* 53 (1996) 743–750, <https://doi.org/10.1103/PhysRevE.53.743>.
- [18] Q. Li, K.H. Luo, Q.J. Kang, Q. Chen, Contact angles in the pseudopotential lattice Boltzmann modeling of wetting, *Phys. Rev. E, Stat. Nonlinear Soft Matter Phys.* 90 (2014) 1, <https://doi.org/10.1103/PhysRevE.90.053301>.

- [19] G. Yan, Z. Li, T. Bore, S.A.G. Torres, A. Scheuermann, L. Li, Discovery of dynamic two-phase flow in porous media using two-dimensional multiphase lattice Boltzmann simulation, *Energies* 14 (2021) 4044, <https://doi.org/10.3390/en14134044>.
- [20] H. Liu, A.J. Valocchi, C. Werth, Q. Kang, M. Oostrom, Pore-scale simulation of liquid CO₂ displacement of water using a two-phase lattice Boltzmann model, *Adv. Water Resour.* 73 (2014) 144–158, <https://doi.org/10.1016/j.advwatres.2014.07.010>.
- [21] H. Huang, J.-J. Huang, X.-Y. Lu, Study of immiscible displacements in porous media using a color-gradient-based multiphase lattice Boltzmann method, *Comput. Fluids* 93 (2014) 164–172, <https://doi.org/10.1016/j.compfluid.2014.01.025>.
- [22] D. Iwahara, H. Shinto, M. Miyahara, K. Higashitani, Liquid drops on homogeneous and chemically heterogeneous surfaces: a two-dimensional lattice Boltzmann study, *Langmuir* 19 (2003) 9086–9093, <https://doi.org/10.1021/la034456g>.
- [23] A.R. Davies, J.L. Summers, M.C.T. Wilson, On a dynamic wetting model for the finite-density multiphase lattice Boltzmann method, *Int. J. Comput. Fluid Dyn.* 20 (2006) 415–425, <https://doi.org/10.1080/10618560601000777>.
- [24] A.G. Yiotis, J. Psihogios, M.E. Kainourgiakis, A. Papaioannou, A.K. Stubos, A lattice Boltzmann study of viscous coupling effects in immiscible two-phase flow in porous media, *Colloids Surf. A, Physicochem. Eng. Asp.* 300 (2007) 35–49, <https://doi.org/10.1016/j.colsurfa.2006.12.045>.
- [25] X. He, S. Chen, R. Zhang, A lattice Boltzmann scheme for incompressible multiphase flow and its application in simulation of Rayleigh–Taylor instability, *J. Comput. Phys.* 152 (1999) 642–663, <https://doi.org/10.1006/jcph.1999.6257>.
- [26] Q. Lou, M. Yang, H. Xu, Wetting boundary condition in an improved lattice Boltzmann method for nonideal gases, *Commun. Comput. Phys.* 23 (2018) 1116–1130, <https://doi.org/10.4208/cicp.oa-2016-0211>.
- [27] A. Hu, L. Li, R. Uddin, D. Liu, Contact angle adjustment in equation-of-state-based pseudopotential model, *Phys. Rev. E* 93 (2016) 053307, <https://doi.org/10.1103/PhysRevE.93.053307>.
- [28] A.J. Briant, P. Papatzacos, J.M. Yeomans, Lattice Boltzmann simulations of contact line motion in a liquid-gas system, *Philos. Trans. R. Soc. A, Math. Phys. Eng. Sci.* 360 (2002) 485–495, <https://doi.org/10.1098/rsta.2001.0943>.
- [29] A.J. Briant, A.J. Wagner, J.M. Yeomans, Lattice Boltzmann simulations of contact line motion. I. Liquid-gas systems, *Phys. Rev. E* 69 (2004) 031602, <https://doi.org/10.1103/PhysRevE.69.031602>.
- [30] T. Lee, L. Liu, Wall boundary conditions in the lattice Boltzmann equation method for nonideal gases, *Phys. Rev. E* 78 (2008) 017702, <https://doi.org/10.1103/PhysRevE.78.017702>.
- [31] H. Kusumaatmaja, E.J. Hemingway, S.M. Fielding, Moving contact line dynamics: from diffuse to sharp interfaces, *J. Fluid Mech.* 788 (2016) 209–227, <https://doi.org/10.1017/jfm.2015.697>.
- [32] H.S. Wiklund, S.B. Lindström, T. Uesaka, Boundary condition considerations in lattice Boltzmann formulations of wetting binary fluids, *Comput. Phys. Commun.* 182 (2011) 2192–2200, <https://doi.org/10.1016/j.cpc.2011.05.019>.
- [33] T. Qian, X.-P. Wang, P. Sheng, Molecular scale contact line hydrodynamics of immiscible flows, *Phys. Rev. E* 68 (2003) 016306, <https://doi.org/10.1103/PhysRevE.68.016306>.
- [34] T. Lee, L. Liu, Lattice Boltzmann simulations of micron-scale drop impact on dry surfaces, *J. Comput. Phys.* 229 (2010) 8045–8063, <https://doi.org/10.1016/j.jcp.2010.07.007>.
- [35] A. Fakhari, D. Bolster, Diffuse interface modeling of three-phase contact line dynamics on curved boundaries: a lattice Boltzmann model for large density and viscosity ratios, *J. Comput. Phys.* 334 (2017) 620–638, <https://doi.org/10.1016/j.jcp.2017.01.025>.
- [36] J.-J. Huang, H. Huang, X. Wang, Wetting boundary conditions in numerical simulation of binary fluids by using phase-field method: some comparative studies and new development, *Int. J. Numer. Methods Fluids* 77 (2015) 123–158, <https://doi.org/10.1002/fld.3975>.
- [37] J.W. Cahn, J.E. Hilliard, Free energy of a nonuniform system. III. Nucleation in a two-component incompressible fluid, *J. Chem. Phys.* 31 (1959) 688–699, <https://doi.org/10.1063/1.1730447>.
- [38] P.-H. Chiu, Y.-T. Lin, A conservative phase field method for solving incompressible two-phase flows, *J. Comput. Phys.* 230 (2011) 185–204, <https://doi.org/10.1016/j.jcp.2010.09.021>.
- [39] Y. Ba, H. Liu, J. Sun, R. Zheng, Color-gradient lattice Boltzmann model for simulating droplet motion with contact-angle hysteresis, *Phys. Rev. E* 88 (2013) 043306, <https://doi.org/10.1103/PhysRevE.88.043306>.
- [40] Z. Xu, H. Liu, A.J. Valocchi, Lattice Boltzmann simulation of immiscible two-phase flow with capillary valve effect in porous media, *Water Resour. Res.* 53 (2017) 3770–3790, <https://doi.org/10.1002/2017WR020373>.
- [41] H. Liu, S. Sun, R. Wu, B. Wei, J. Hou, Pore-scale modeling of spontaneous imbibition in porous media using the lattice Boltzmann method, *Water Resour. Res.* 57 (2021) 1–24, <https://doi.org/10.1029/2020WR029219>.
- [42] L. Wang, H. Huang, X.-Y. Lu, Scheme for contact angle and its hysteresis in a multiphase lattice Boltzmann method, *Phys. Rev. E* 87 (2013) 013301, <https://doi.org/10.1103/PhysRevE.87.013301>.
- [43] J.-J. Huang, H. Huang, X. Wang, Numerical study of drop motion on a surface with stepwise wettability gradient and contact angle hysteresis, *Phys. Fluids* 26 (2014) 062101, <https://doi.org/10.1063/1.4880656>.
- [44] A. Fakhari, M. Geier, D. Bolster, A simple phase-field model for interface tracking in three dimensions, *Comput. Math. Appl.* 78 (2019) 1154–1165, <https://doi.org/10.1016/j.camwa.2016.08.021>.
- [45] H. Liang, H. Liu, Z. Chai, B. Shi, Lattice Boltzmann method for contact-line motion of binary fluids with high density ratio, *Phys. Rev. E* 99 (2019) 063306, <https://doi.org/10.1103/PhysRevE.99.063306>.
- [46] A. Begmohammadi, R. Haghani-Hassan-Abadi, A. Fakhari, D. Bolster, Study of phase-field lattice Boltzmann models based on the conservative Allen-Cahn equation, *Phys. Rev. E* 102 (2020) 023305, <https://doi.org/10.1103/PhysRevE.102.023305>.
- [47] H. Liu, Y. Ju, N. Wang, G. Xi, Y. Zhang, Lattice Boltzmann modeling of contact angle and its hysteresis in two-phase flow with large viscosity difference, *Phys. Rev. E* 92 (2015) 033306, <https://doi.org/10.1103/PhysRevE.92.033306>.
- [48] Y.H. Qian, D. D’Humières, P. Lallemand, Lattice BGK models for Navier-Stokes equation, *Europhys. Lett.* 17 (1992) 479–484, <https://doi.org/10.1209/0295-5075/17/6/001>.
- [49] P. Lallemand, L.-S. Luo, Theory of the lattice Boltzmann method: dispersion, dissipation, isotropy, Galilean invariance, and stability, *Phys. Rev. E* 61 (2000) 6546–6562, <https://doi.org/10.1103/PhysRevE.61.6546>.
- [50] R. Huang, H. Wu, A modified multiple-relaxation-time lattice Boltzmann model for convection–diffusion equation, *J. Comput. Phys.* 274 (2014) 50–63, <https://doi.org/10.1016/j.jcp.2014.05.041>.
- [51] R. Huang, H. Wu, N.A. Adams, Lattice Boltzmann model with self-tuning equation of state for multiphase flows, *Phys. Rev. E* 99 (2019) 023303, <https://doi.org/10.1103/PhysRevE.99.023303>.
- [52] L. Fei, J. Du, K.H. Luo, S. Succi, M. Lauricella, A. Montessori, Q. Wang, Modeling realistic multiphase flows using a non-orthogonal multiple-relaxation-time lattice Boltzmann method, *Phys. Fluids* 31 (2019) 042105, <https://doi.org/10.1063/1.5087266>.
- [53] F. Ren, B. Song, M.C. Sukop, H. Hu, Improved lattice Boltzmann modeling of binary flow based on the conservative Allen-Cahn equation, *Phys. Rev. E* 94 (2016) 023311, <https://doi.org/10.1103/PhysRevE.94.023311>.
- [54] H.L. Wang, Z.H. Chai, B.C. Shi, H. Liang, Comparative study of the lattice Boltzmann models for Allen-Cahn and Cahn-Hilliard equations, *Phys. Rev. E* 94 (2016) 1–7, <https://doi.org/10.1103/PhysRevE.94.033304>.
- [55] A. Fakhari, T. Mitchell, C. Leonardi, D. Bolster, Improved locality of the phase-field lattice-Boltzmann model for immiscible fluids at high density ratios, *Phys. Rev. E* 96 (2017) 053301, <https://doi.org/10.1103/PhysRevE.96.053301>.
- [56] X. Meng, Z. Guo, Multiple-relaxation-time lattice Boltzmann model for incompressible miscible flow with large viscosity ratio and high Péclet number, *Phys. Rev. E* 92 (2015) 043305, <https://doi.org/10.1103/PhysRevE.92.043305>.
- [57] T. Lei, X. Meng, Z. Guo, Pore-scale study on reactive mixing of miscible solutions with viscous fingering in porous media, *Comput. Fluids* 155 (2017) 146–160, <https://doi.org/10.1016/j.compfluid.2016.09.015>.
- [58] F. Diotallevi, L. Biferale, S. Chibbaro, A. Lamura, G. Pontrelli, M. Sbragaglia, S. Succi, F. Toschi, Capillary filling using lattice Boltzmann equations: the case of multi-phase flows, *Eur. Phys. J. Spec. Top.* 166 (2009) 111–116, <https://doi.org/10.1140/epjst/e2009-00889-7>.
- [59] T. Lee, C.-L. Lin, A stable discretization of the lattice Boltzmann equation for simulation of incompressible two-phase flows at high density ratio, *J. Comput. Phys.* 206 (2005) 16–47, <https://doi.org/10.1016/j.jcp.2004.12.001>.